

# Data Collection

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PRI 26/27 · Information Processing and Retrieval

M.EIC · Master in Informatics Engineering and Computation

M.IA · Master in Artificial Intelligence

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# Data Processing Overview

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## Data **C**ollection

Sources, formats  
Ingestion methods  
Metadata basics

## Data **P**reparation

Cleaning, transformation  
Quality and integration  
Exploratory analysis

## Data **P**rocessing

Storage models  
Batch / stream pipelines  
Preparation & visualization

# Today's Plan

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- Context and motivation for data processing
- Focus on data collection and preparation
  - Data collection and preparation tasks
  - Overview of a typical pipeline
  - Example projects
- Review of the first milestone delivery
- Review the plan for the next practical class

# Overview

# "Data"

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- In Latin, data is the plural of datum.
- In specialized fields, data is treated as plural, e.g. "data were collected".
- Generally, it is treated as a mass noun, like "information", e.g. "data was collected".
- We adopt the use of "data" as mass noun.

# Terminology: Data, Metadata and Information

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- **Data**

- is a measurement of something on a scale;
- a fact known by direct observation.

- **Metadata**

- is "data about data";
- not the content of data but data providing information about one or more aspects of the data, such as description (date, time, author), structure (format, version), administrative (permissions), legal, etc.

- **Information**

- is data with a context / meaning, thus enabling decision making;
- is data that has been processed, organized and structured.

# Terminology: Decimal and Binary Systems

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- The **binary system uses powers of 2**.
- The **decimal system uses powers of 10**.
- In the International System of Units standard, kilo, mega, giga, correspond to powers of 1000 — thus decimal prefixes.
- Historically, the computer industry used the same prefix with two different meanings, i.e. 1 MB could either be 1 048 576 bytes (binary) or 1 000 000 bytes (decimal).
- In 2008, binary prefixes — i.e., that refer to powers of 2 — were officially introduced: kibi (Ki), mebi (Mi), gibi (Gi), tebi (Ti), pedi (Pi), exbi (Ei), zebi (Zi), yobi (Yi).
  - 1 MiB ( $2^{20} = 1024^2$ ) = 1 048 576 bytes
  - 1 MB ( $1000^2$ ) = 1 000 000 bytes

# Out of Scope

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- Outside the scope of this course are:
  - Ethical dimensions of data and information
  - Economic aspects of data
  - Legal aspects
  - ....
- Note that these concepts should not be foreign to informatics or computer science.
- Informatics engineers need to be aware of many of these aspects, particular those with strong social impact, in their work. There are many recent examples, e.g. social media, cloud services.
- Possible topics for invited talks.

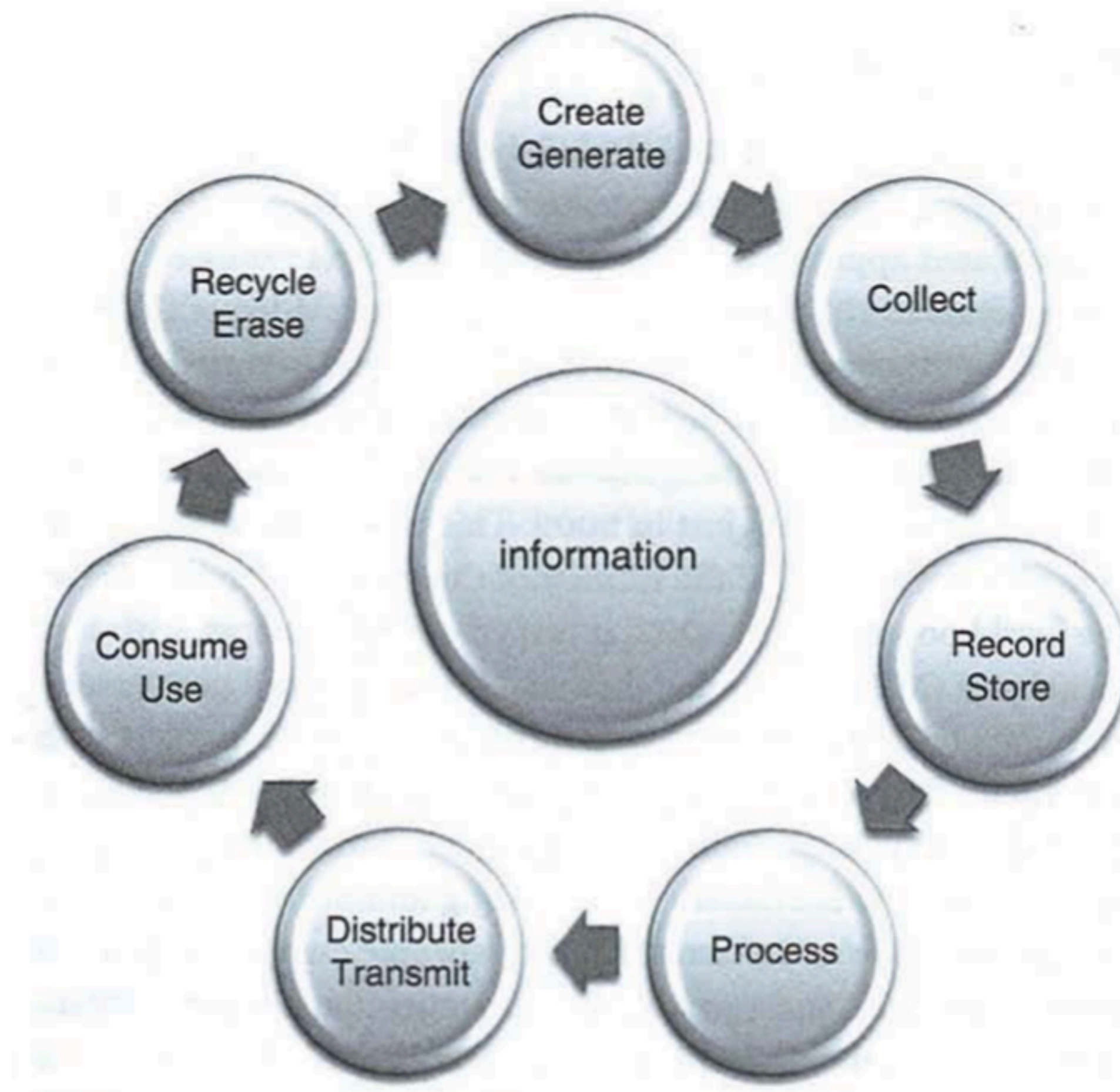
# Information Life Cycle

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- In modern societies, progress and welfare is increasingly dependent on the successful and efficient management of the life cycle of information.
- The life cycle of information typically includes the following phases:
  - Occurrence: discover, design, author, etc;
  - Transmission: networking, accessing, retrieving, transmitting, etc;
  - Processing and Management: collecting, validating, modifying, indexing, classifying, filtering, sorting, storing, etc;
  - Usage: monitoring, explaining, planning, forecasting, decision-making, educating, learning, etc;
- Information and Communication Technologies (ICT) evolved from being mainly recording systems, to being communication systems, to also (and currently) being processing and producing systems.

# Information Life Cycle

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# The zettabyte\* era

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- A study from 2003, reported that humanity had accumulated approximately 12 exabytes of data until the emergence of the personal computer.
- Of these, 92% were stored on magnetic media (i.e. digital media).
- A more recent study from 2018, reported that *"the total amount of data created, captured, copied and consumed in the world was 33 zettabytes (ZB) – the equivalent of 33 trillion gigabytes. This grew to 59 ZB in 2020 and is predicted to reach a mind-boggling 175 ZB by 2025. One zettabyte is 8,000,000,000,000,000,000,000 bits"* \*\*
- Also, *"there are around 600 hyperscale data centres – ones with over 5,000 servers – in the world. Around 39% of them are in the US, while China, Japan, UK, Germany and Australia account for about 30% of the total"* \*\*
- With these trends, i.e. an annual growth rate of 50%, ~150 years from now the number of bits would surpass the number of atoms on Earth ...

\* 1 zettabyte = 1000 exabytes; 1 exabyte = 1000 petabytes; 1 petabyte = 1000 terabytes.

\*\* Vopson, M. *The world's data explained: how much we're producing and where it's all stored. The Conversation*, 2021  
<https://theconversation.com/the-worlds-data-explained-how-much-were-producing-and-where-its-all-stored-159964>

# Value in Data

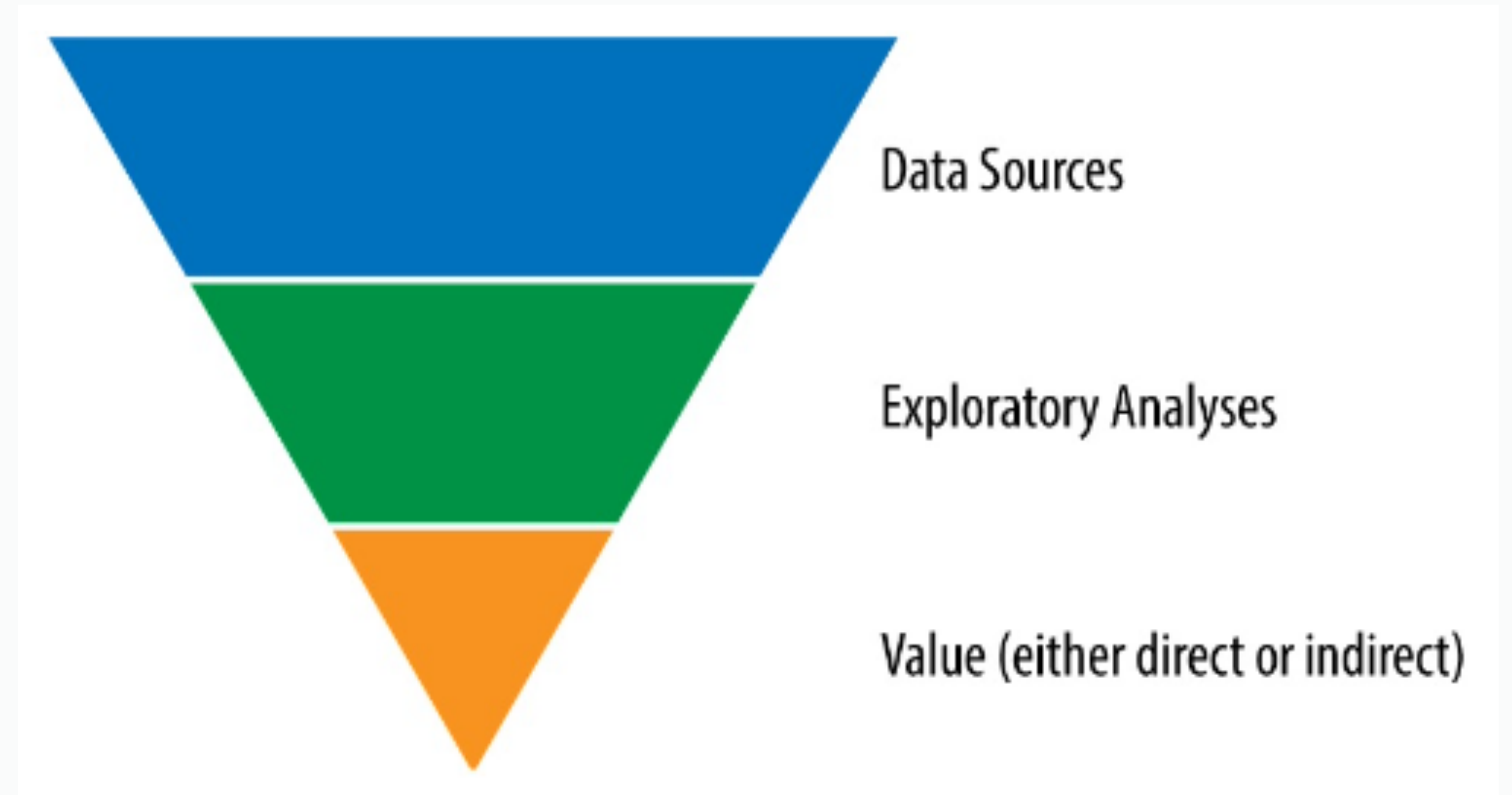
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- "Data is the new oil", *everybody* (circa 20xx).
- Data is a source of value generation, providing evidence and content for the design of new products, new processes, and contributions to more efficient operations.
- In data-driven approaches, multidisciplinary teams experiment and explore large and diverse sources of data to "extract signal from the noise".
- **Indirect value** — data provides value by influencing or supporting decisions, e.g. risk analysis in insurance, purchase decisions in retail.
- **Direct value** — data provides value by feeding automated systems, e.g. search systems, product recommendation systems.

# Data Value Funnel

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- A large number of data sources and exploratory analysis are required to produce a single valuable application of data.
- Minimize the time spent on non-relevant data by empowering business-experts (i.e. people who know about the business) to explore data.
- Additionally, make data analysis processes as efficient as possible, e.g. by implementing effective data processing workflows.



# Increasing Data Value

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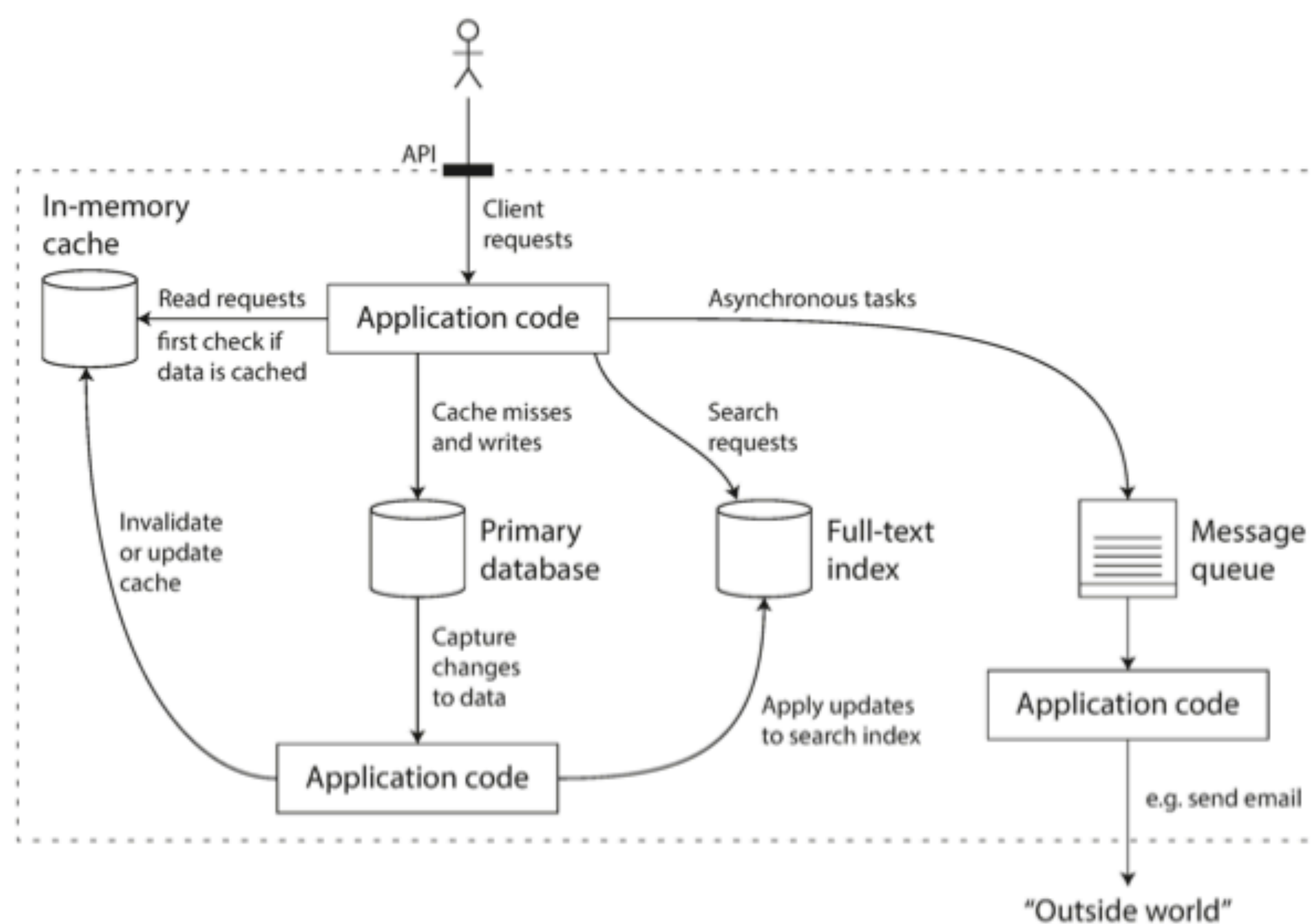
- Make data available, i.e. simply make previously inaccessible data available.
- **Combine** data, i.e. create a single coherent whole from disparate data sources; e.g. collection of news from the main news outlets during a year.
- **Clean** data, i.e. eliminate problems such as incomplete values, duplicates; or create a subset according to specific criteria.
- **Structure** data, i.e. provide structure to unstructured data; e.g. derive entity mentions field from a textual field.
- **Enrich** data, i.e. complement existing data with data from other sources, including the computation necessary to do so.

# Data-Intensive Applications

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- Many applications today are data-intensive, as opposed to compute-intensive.
- In this context, existing problems are:
  - the **volume** of data available;
  - the **complexity** of the data;
  - the **speed** at which it changes.
- Common building blocks in data-intensive applications include:
  - Store data, for use or sharing (databases);
  - Remember the result of an expensive operation (caches);
  - Enable search and filtering (indexes)
  - Send messages between systems (stream processing);
  - Periodically process large amounts of data (batch processing);

# Example of a Data System



# Data Stages

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- Data moves through three main stages:
  - **Raw** — focus is on data discovery; the primary goals are ingestion, understanding, and metadata creation; common questions include: what kinds of records are in the data? how are record fields encoded?
  - **Refined** — focus is on data preparation for further exploration; tasks include removing unwanted parts, reshaping poorly formatted elements; establishing relationships between datasets; assessing data quality issues.
  - **Production** — focus is on integrating the data into production processes or products.
- Several data processing patterns are described in the literature: ETL, ELT, OSEMN.

# ETL Pattern

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- The ETL framework (extract-transform-load) was coined in the 1970s and popularized in the context of data warehousing.
  - **Extract**, involves extracting data from the source system.
  - **Transform**, a series of operations or transformations are applied to the extracted data.
  - **Load**, involves publishing data to the target system, either simple flat files or other infrastructures.
- ETL is usually associated with classic centralized IT driven operations.
- Still relevant in cloud environments as well.

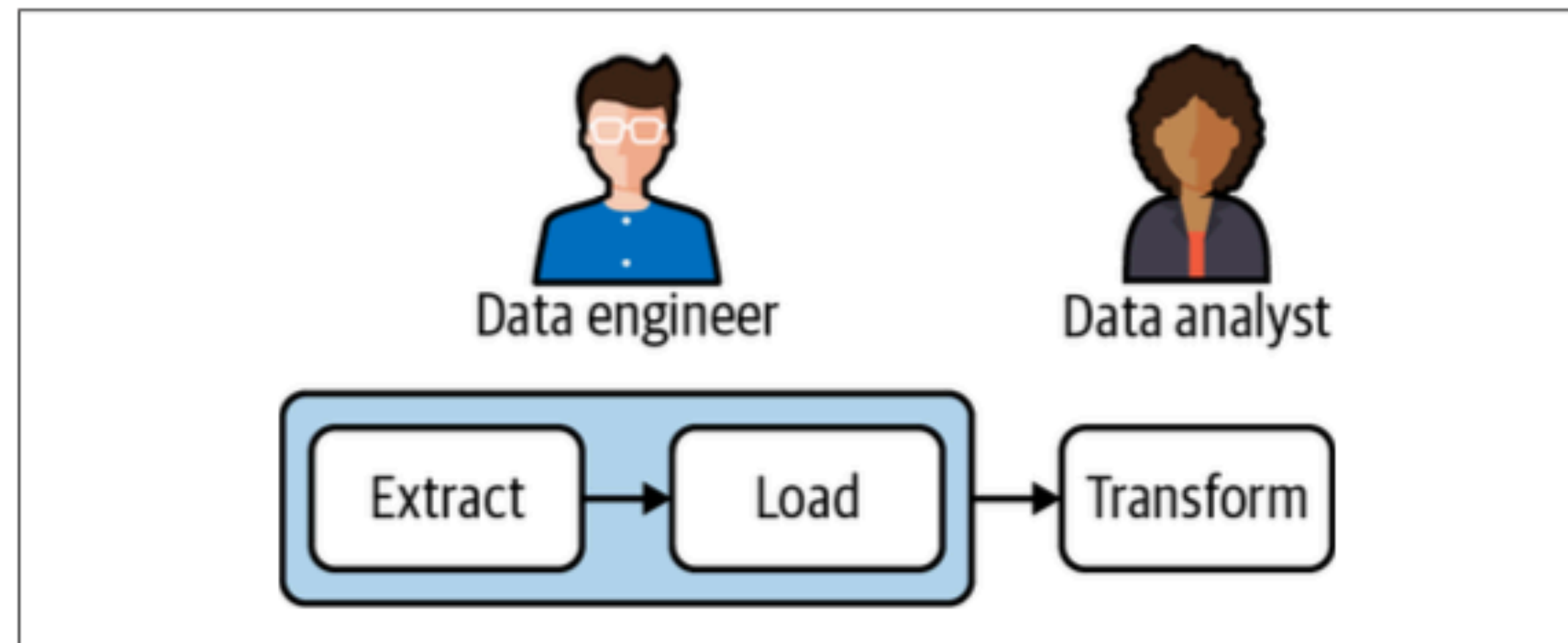
# ELT and EtLT Frameworks

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- ELT (**extract-load-transform**) is a recent evolution over the ETL framework.
- Load-transform, in contrast with transform-load, is a pattern more well-suited to the division of responsibilities in multidisciplinary teams.
- Increasingly common access to data storage infrastructures capable of handling large volumes of data has led to a more flexible pattern.
- Column-oriented data structures are particularly well-suited to typical data processing tasks, i.e. organizing operations per field or property.
- The sub-pattern EtLT introduces a transformation step before the loading, typically associated with data cleaning tasks.

# ELT Pattern

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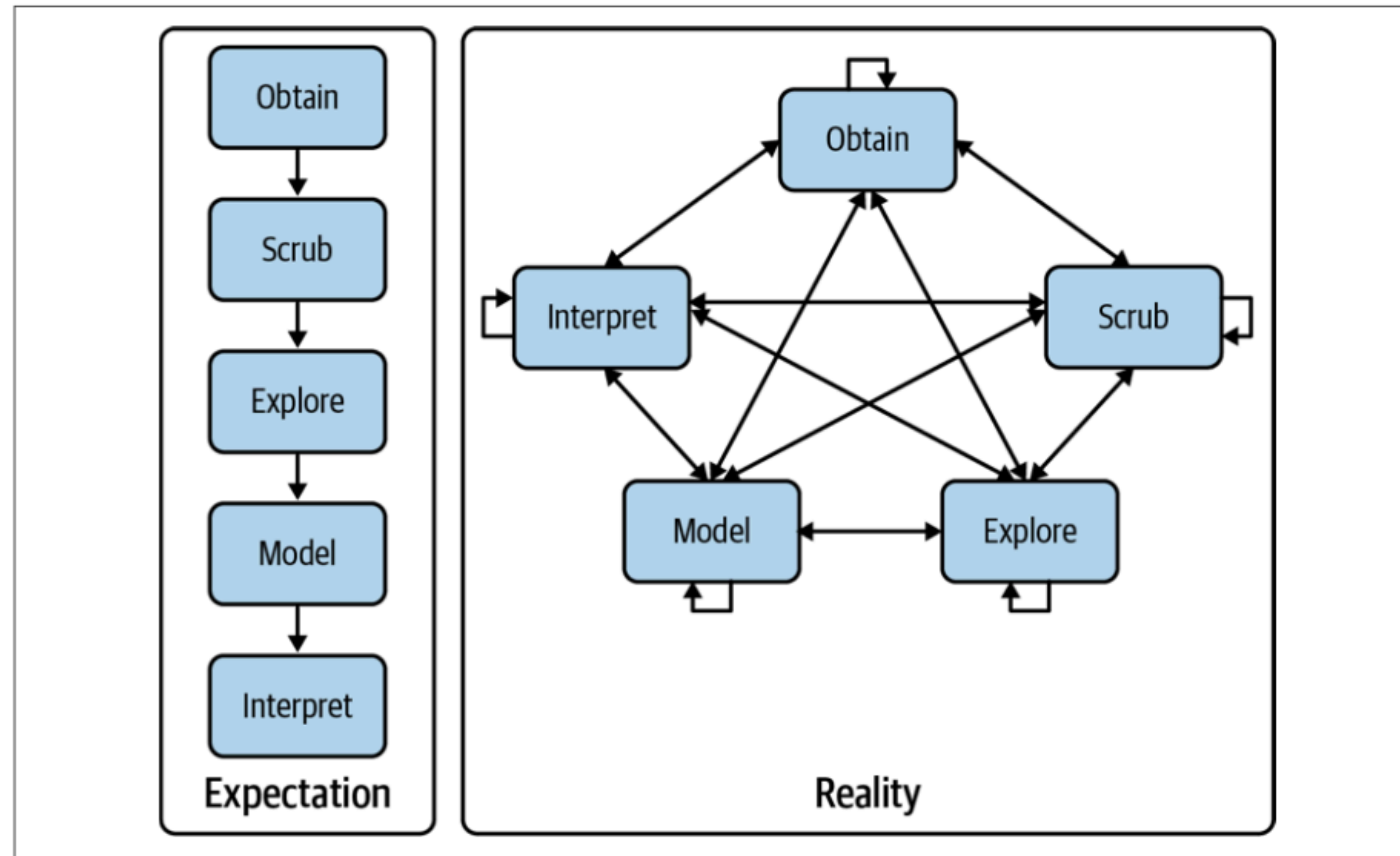
*Figure 3-3. The ELT pattern allows for a clean split of responsibilities between data engineers and data analysts (or data scientists). Each role can work autonomously with the tools and languages they are comfortable in.*

# OSEMN Framework

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- In the context of Data Science, the OSEMN (“awesome”) acronym was coined.
  - **Obtain**, gathering data.
  - **Scrub**, clear, arrange, prepare data.
  - **Explore**, observe, experiment, visualize.
  - **Model**, create a statistical model of the data.
  - **Interpret**, draw conclusions, evaluating and communicating results.
- Although presented as a series of steps, real-world processes are typically non-linear.

# Iterative Process



*Figure 1-1. Doing data science is an iterative and nonlinear process*

# Data Analytics Process in Organizations

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- Data analytics processes can either be centralized, typically in an IT department, or decentralized, in specialized teams.
- Benefits of centrally controlling data processing:
  - Controlled data governance;
  - Efficiency gains due to reuse of practices, methods, and expertise.
- Drawbacks of centralization:
  - Frequent bottlenecks due to dependency on the IT department.
- Challenge: expanding the range of users who have access to raw data and provide them with the necessary training and skills.

# Data Engineers

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- Data engineers have emerged as an autonomous key role in this context.
- Design, implement and maintain data processing pipelines.
- Work closely with data scientists and analysts to understand what will be done with the data.
- Wide range of technical skills:
  - SQL and Data Warehousing
  - Programming - Python, Java, Go (common in this context)
  - Distributed Computing
  - Cloud Infrastructures
  - System Administration

# Data Engineering Lifecycle

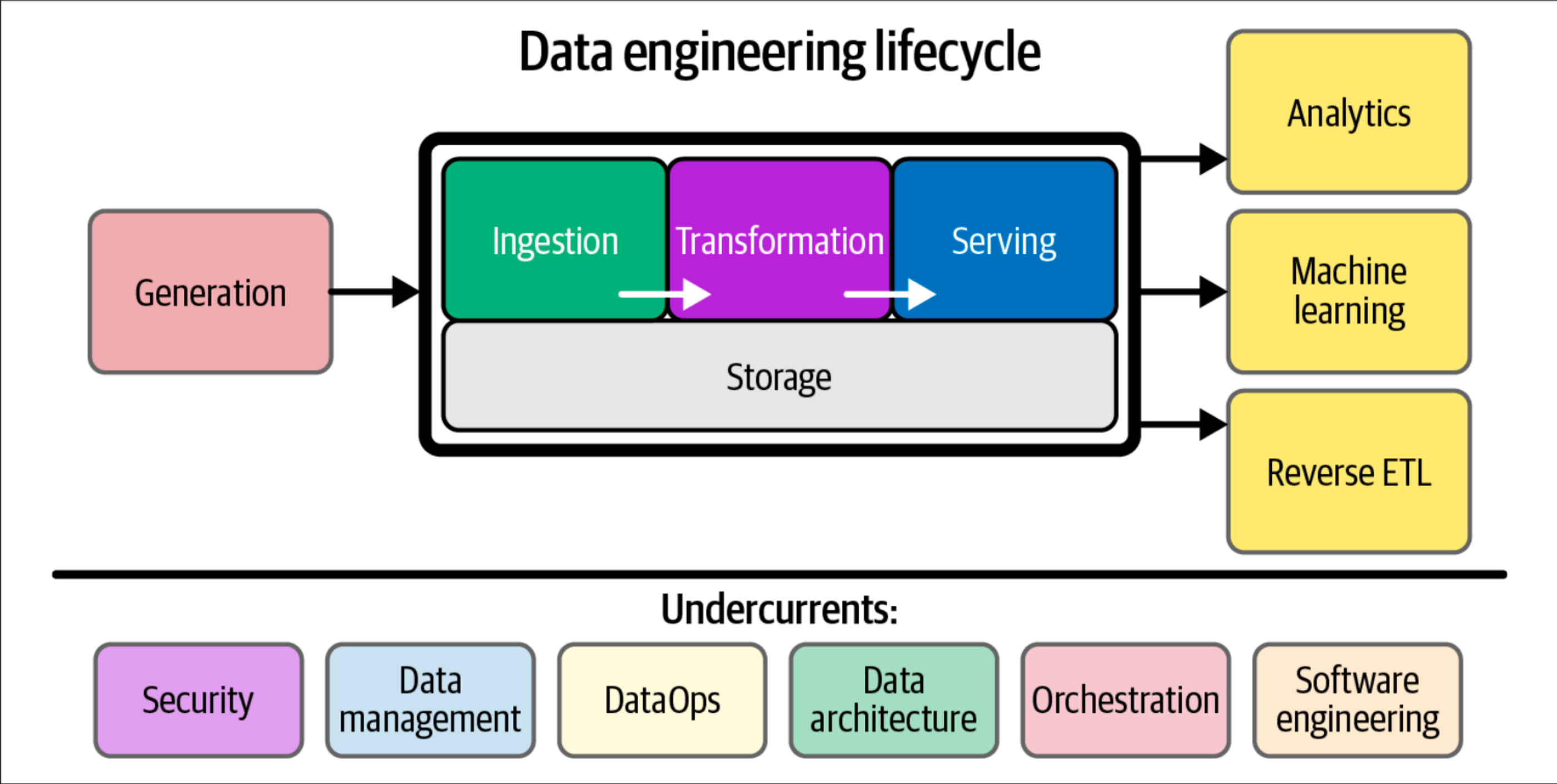


Figure 2-1. Components and undercurrents of the data engineering lifecycle

# Data Collection

# Diversity of Data Sources

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- Data sources vary across many dimensions:
  - **Ownership** — either owned or from third-parties; understanding ownership is central, i.e. know what data you have access to and what you can do with it;
  - **Ingestion interface and structure** — how do you get the data and in what form it is provided;
  - **Volume** —in each step of the pipeline, volume needs to be taken into account; high- and low- volume are difficult to define and depend on available infrastructures and algorithms;
  - **Cleanliness and validity** —duplicate data, missing or incomplete data, encoding, etc;
  - **Latency and bandwidth of the source** — need to consider internal update requirements and also source system limits, speed, timeouts, etc.

# Open Data

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- The idea that data should be freely available to anyone, to use, modify, and republish for any purpose.
- Associated with a movement, see Open Knowledge Foundation, <https://okfn.org/>
- One of the most important forms of open data is open government data.
- Open data can also be linked, known as linked open data (LOD), see <https://www.w3.org/DesignIssues/LinkedData.html> (2009)
- "Web of Data" is an expression coined to represent the set of technologies and practices that enable a space where data can be automatically discovered and accessed by machines.
- Also related is the concept of FAIR: findable, accessible, interoperable, and reusable; emphasizing machine-actionability over data.
- FAIR/O is used to indicate that a data source complies with FAIR and is also of open nature.

# Data Sources (Examples)

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- Search
  - Google Dataset Search, <https://toolbox.google.com/datasetsearch>
- Structured
  - dados.gov, <https://dados.gov.pt>
  - Dados Abertos do Parlamento, <https://www.parlamento.pt/Cidadania/Paginas/DadosAbertos.aspx>
- Unstructured
  - Legislação Portuguesa Consolidada, <https://dre.pt/web/guest/legislacao-consolidada>
  - Acórdãos do Tribunal Constitucional, <http://www.tribunalconstitucional.pt/tc/acordaos/>
- *See Moodle for a list of selected data sources and example datasets.*

# Data Selection - Things to Consider

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- Is the author a trustworthy source that can be contacted?
- Is the data regularly updated?
- Does the data include information about how and when it was acquired?
- Does it seem plausible through observation?

# Ingestion Interfaces and Data Structures

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- Examples of **ingestion interfaces** include:
  - A relational database behind an application, such as PostgreSQL, SQLite or Oracle;
  - A layer of abstraction on top of a system, such as a REST API;
  - An endpoint to a message queue system, such as RabbitMQ;
  - A shared network filesystem, containing logs, CSV files, or other flat files.
- Examples of **data structures** include:
  - JSON from REST API;
  - Well-structured data from a relational database;
  - Semistructured log data;
  - CSV (comma-separated values) datasets;
  - PDF, or other proprietary format, files;
  - HTML, or other semi-structured, files;

# Data Formats

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- Data formats enable the representation of different types of data in a computer-usable form.
- Common data representations:
  - Alphanumeric: Unicode, ASCII
  - Image (bitmap): PNG, JPG
  - Image (object / vector): PostScript, SVG
  - Sound: AVI, MP3, AAC
  - Documents: PDF, HTML, XML
- Formats used by individual applications are known as **proprietary formats**.
- Proprietary formats can be open if its specifications are published. Although not always free of licensing restrictions.
- Some proprietary formats become "proprietary standards" when they become *de facto* standards due to general use.
- An **open format** is defined by a published specification, usually maintained by a standards organization (e.g., PNG, FLAC).

# Data Encoding

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- Data can be encoded
  - in memory, in specific structures such as objects, lists, arrays;
  - as a self-contained sequence of bytes, for file storage or network transmission, e.g. JSON document.
- The process of translating from the in-memory representation to a byte sequence is called **encoding** (also known as serialization), and the inverse is called **decoding** (also parsing, deserialization).
- Most programming languages have built-in support for encoding (and decoding) in-memory data to byte sequences, e.g. Java's `java.io.Serializable`, Python's `pickle`, PHP's `serialize`.
- Useful for transient cases, but in general not recommended in production pipelines — limited to a programming language, reduced interoperability, lower performance, etc.

# JSON, XML Serialization

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- JSON and XML are the most common text-based encoding standards.
- JSON is widely supported by many applications.
- CSV is also popular, but less powerful (no internal structure, no explicit types).
- These are (somewhat) human readable (an advantage).
- Are the *de facto* solution for data interchange, between organization, between applications, as export formats in applications, as API outputs.
- Limitations include: ambiguous support for number formats, limited support for binary data (e.g. images).

# Binary Serialization

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- Binary serialization is more compact and faster to parse.
- Is a common solution for within organization data exchange.
- There are many binary formats for JSON - MessagePack, BSON, BSON, etc.
- Apache Thrift (originally Facebook) and Protocol Buffers (protobuf) are open-source binary encoding libraries that produce a binary encoding of a given record.

# Columnar Formats

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- Designed for analytical processing of large datasets.
- Store data in a column-oriented layout, efficient for queries on subsets of columns.
- Support compression and encoding per column: smaller files, faster scans.
- Commonly adopted in data lakes and big data tools (Spark, Hive, DuckDB, Pandas, BigQuery).
- **Parquet**, <https://parquet.apache.org>
- **ORC** (Optimized Row Columnar), <https://hive.apache.org/docs/latest/language/languagemanual-orc>
- **Avro**, <https://avro.apache.org>

# Data Quality

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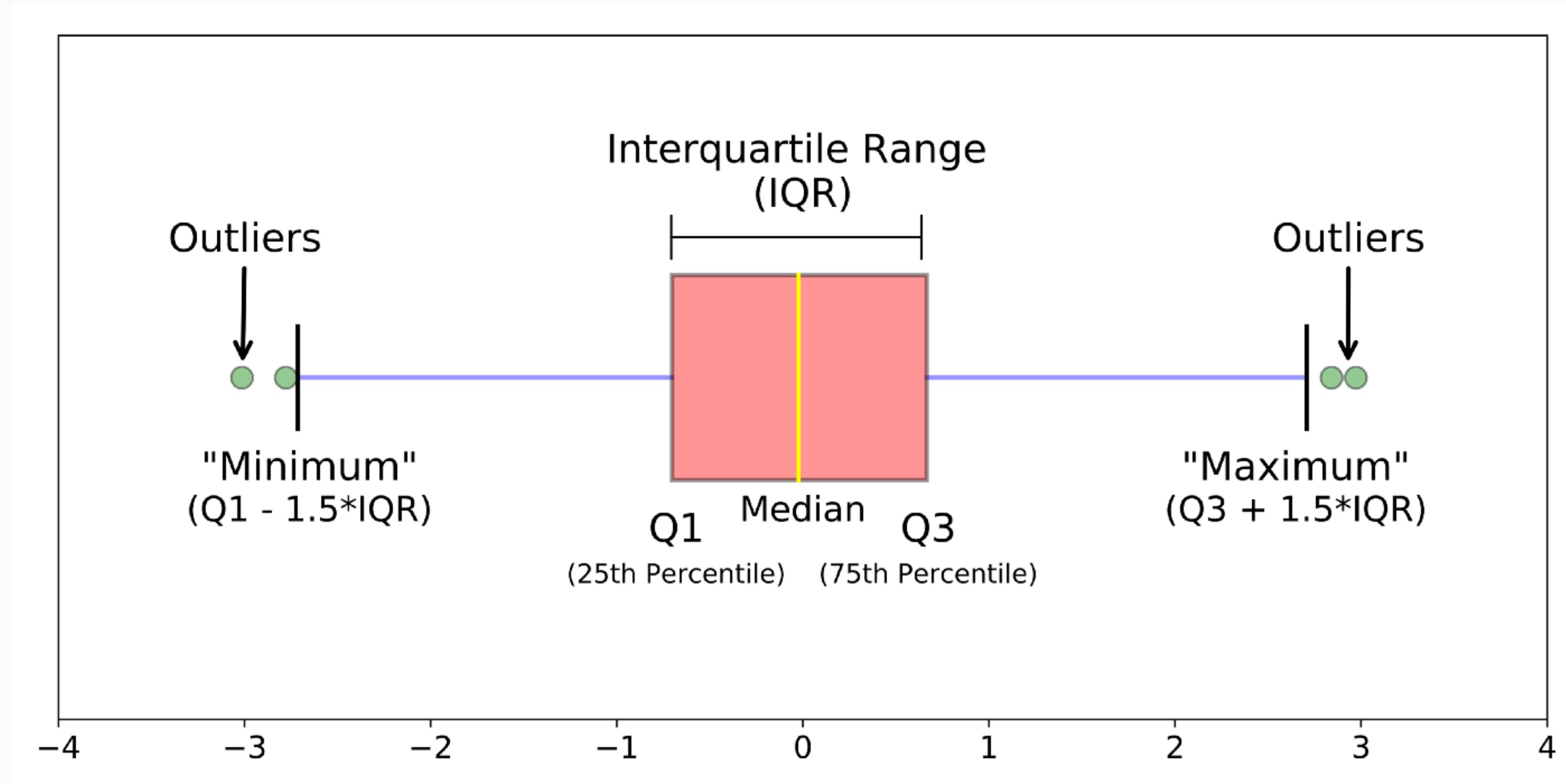
- Common problem affecting data quality:
  - **Missing data** — due to error, specific meaning (e.g., n/a, 0, NULL).
  - **Inconsistent values** — distinct timezones in time/dates, multiple units (e.g., m, km).
  - **Precision problems** — rounding decisions may result in fake patterns (e.g., maps).
  - **Duplicate values** — due to errors, or valid data.
  - **Other common issues** — text encoding problems, mislabeled data, incomplete, outdated, etc.
- There is a need to investigate and understand data properties during the data selection phase — often called data investigation or assessment.

# Overall View of the Distribution

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- Descriptive statistics are commonly used to begin the investigation.
- Common measures and techniques estimate the:
  - **Central tendency**, i.e. central, average value of observations:
    - Mean (sum divided by the number of items), median (middle value that separates the observations), mode (most frequent value);
  - **Dispersion**, i.e. how much the values vary:
    - Standard deviation, interquartile range (difference between values in upper and lower quartile), difference between the maximum and minimum.
- **Box plots** are methods that graphically depict most of these descriptive statistics.

# Box plots

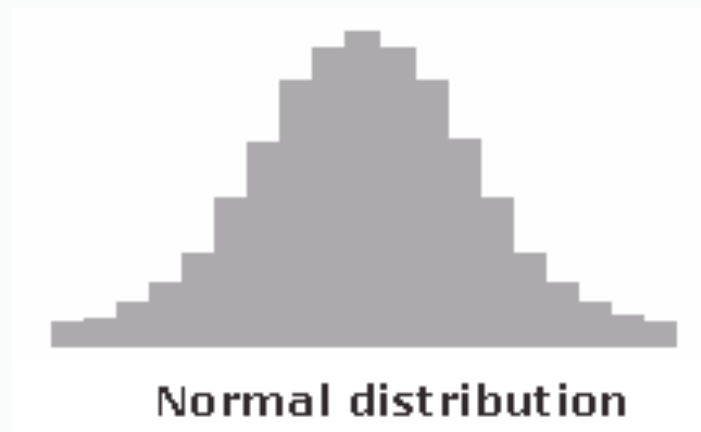


- Image from Understanding Boxplots, Towards Data Science  
[ <https://towardsdatascience.com/understanding-boxplots-5e2df7bcbd51> ]

# Frequency Histograms

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- For large data volumes, observing distributions of attribute values can be done using histograms, which represent how numerical data is distributed.
- A key aspect of producing histograms is exploring with different bin sizes.
- Numerous distributions exist and are described in detail in statistical literature.

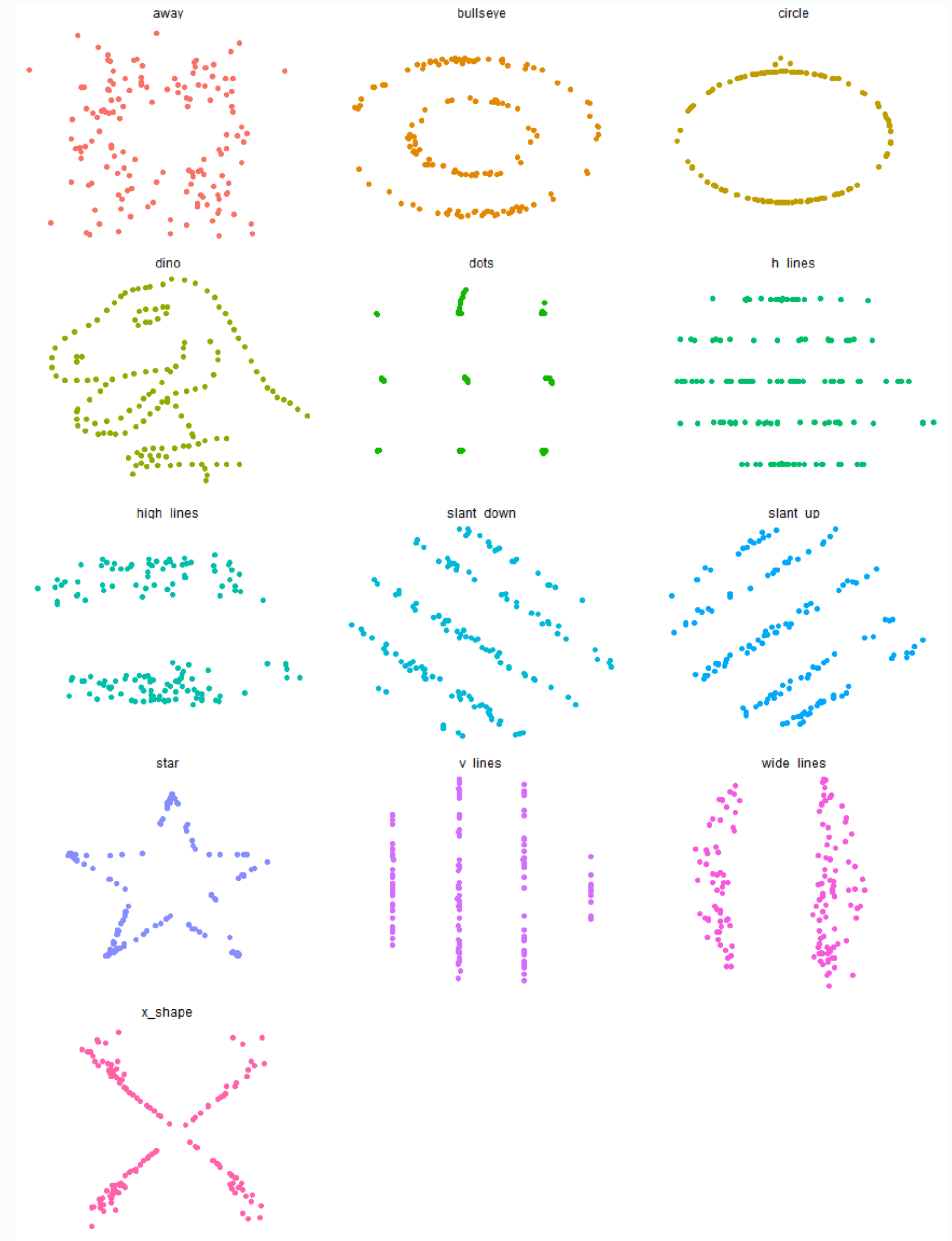


- Examples from "What are Histograms?", ASQ  
[ <https://asq.org/quality-resources/histogram> ]



# Descriptive statistics only provide a summary

- The image on the right depicts 13 sets of x-y data, where basic descriptive statistics have the same values, i.e. x-mean, y-mean, x-std, y-std), but look very different.
- Don't rely only on descriptive statistics, include exploratory visualization in your process.
- Image from The Datasaurus data package [ <https://cran.r-project.org/web/packages/datasauRus/vignettes/Datasaurus.html> ]



# Outliers

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- Outliers are items that differ significantly from others.
- It is necessary to understand if these values are exception but valid cases, or if they are errors that need to be removed. Expert domain-knowledge is often necessary.
- Errors resulting in outliers may be the result of:
  - Problems in the data collection procedure;
  - Hardware or software problems in data collection tools;
  - Human mistakes in data recording.
- Outliers may significantly distort descriptive statistics or visualizations.

# Missing Data

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- Missing data is an important aspect of data quality that always needs a detailed investigation to determine its origin and impact on following steps.
- Isolated instances of missing data aren't usually a problem, however if the missing data is not randomly distributed or occurs in large numbers globally or in specific variables, the data set will be biased and thus not appropriate for a valid analysis.
- Missing data can also be an indicator of flaws in the data collection process.

# Data Quality Summary

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- Investigating the properties of the data at its origin and at different points of the data processing pipeline is a key aspect of a solid data-based project, helping on:
  - Deciding on the data sources to select;
  - Determining possible bias and limitations of the data *a priori*;
  - Detecting and correcting problems in the pipeline;
  - Framing the conclusions of built products;
- Data quality investigations should rely on multiple methods, from descriptive statistics to exploratory visualization.
- Best practices: clean and validate in the best system to do so; validate often.

# Tools for Data Collection

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- SQL for extracting data from databases.
- Custom code for APIs.
- Unix commands to work with external sources, e.g. curl, wget.
- Web crawling platforms:
  - Scrapy, <https://scrapy.org/> [Python]
  - Internet Archive Heritrix, <https://github.com/internetarchive/heritrix3> [Java]
  - Apache Nutch, <http://nutch.apache.org> [Java]

# Covered Today

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- The role and **value of data** in modern systems
- Key **terminology**: data, metadata, information, and decimal vs. binary prefixes
- The **information life cycle** and data stages (raw → refined → production)
- Main **data processing frameworks**: ETL, ELT, EtLT, OSEMN
- The role of **data engineers** and the data engineering lifecycle
- Diversity of **data sources, interfaces, and formats**
- Importance of **data quality** (missing data, duplicates, outliers)
- Use of **exploratory analysis** (descriptive stats, box plots, histograms)

# Bibliography and Further Reading

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**Questions or comments?**